

CO₂ AT-1

1) suppose that the data for Analysis include the attribute age. The age values for data types are (in increasing order) 13, 16, 16, 19, 20, 20, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

a) what is the mean of the Data?

what is the median.

$$\text{Mean} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$= \frac{809}{27} \rightarrow 29.96 \approx 30$$

$$\text{Median} = \frac{n+1}{2}$$

$$= \frac{27+1}{2} = 14.$$

b) what is the Mode of Data? and Modality.

$$\text{Mode} = 25, 35.$$

25 and 35 are repeating four times. so it is Bimodal.

c) what is the Midrange of the Data?

$$\text{Midrange} = \frac{\text{max} + \text{min}}{2} = \frac{70 + 13}{2}$$

$$= 41.5$$

d) Can you Find the First Quartile and the third Quartile (Q_3) of the Data ~~Find~~

$$Q_1 = 20$$

$$Q_3 = 35$$

$$IQR = Q_3 - Q_1$$

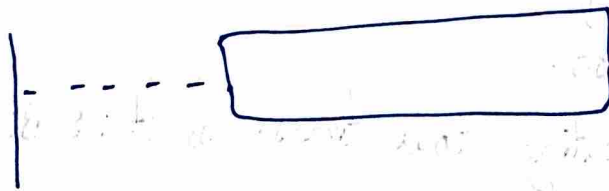
$$= 35 - 20$$

$$= 15$$

e) Give the five-Number summary of the Data.

<u>Metric</u>	<u>Value</u>
Min	13
Q_1	20
Q_2	25
Q_3	35
Max	40

f) Boxplot of the Data.



13 15 16 19 20 21 22 25 30 33 35 36 40

45:

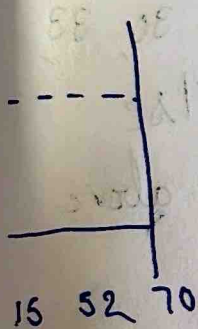
2) Given the Following measurements for the variables are 18, 22, 25, 24, 42, 28, 43, 33, 56, 28 standardize the variable by the following:-

Compute the Mean Absolute Deviation of age.

$$= \frac{18 + 22 + 25 + 24 + 42 + 28 + 43 + 33 + 56 + 28}{10}$$

$$= \frac{330}{10} = 33$$

Mean Absolute Deviation:



Age
(n)

$|x - \bar{x}|$

18

18 - 33

22

22 - 33

25

25 - 33

42

42 - 33

28

28 - 33

43

43 - 33

33

33 - 33

5

10

0

35

35 - 33

2

56

56 - 33

23

26

26 - 33

5

sum of Absolute deviation:

$$15 + 11 + 8 + 9 + 5 + 10 + 0 + 2 + 23 + 5 = 88$$

$$= \frac{88}{10} \rightarrow 8.8$$

3) Suppose the Data for Analysis Includes the attribute age. The age values for the Data types are 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70. Else smoothing by him means to smooth the above Data, using a bin depth of 3.

step 1:

Total No. of values = 27

Bins Depth = 3

Total No. of Bins $\rightarrow \frac{\text{Total}}{\text{Bin depth}}$

$$= \frac{27}{3} \rightarrow 9$$

<u>Bin</u>	<u>original</u> <u>Values</u>	<u>Big</u> <u>mean</u>	<u>Smoothed</u> <u>Values</u>
		14.67	14.67, 14.67, 14.67
B ₁	13, 15, 16		
B ₂	16, 19, 20	18.33	18.33, 18.33, 18.33
B ₃	20, 21, 22	21.00	21.00, 21.00, 21.00
		24.00	24.00, 24.00, 24.00
B ₄	22, 25, 30		
		26.67	26.67, 26.67, 26.67
B ₅	25, 25, 30		
		33.67	33.67, 33.67, 33.67
B ₆	33, 33, 35		
		25.00	25.00, 25.00, 25.00
B ₇	35, 35, 35		
		40.00	40.00, 40.00, 40.00
B ₈	36, 40, 45		
		56.00	56.00, 56.00, 56.00
B ₉	46, 52, 70		

4) use the Two methods below to Normalize the following group of Data : 200, 300, 400, 600, 1000(a)

min-max normalization by setting min = 0 and max = 1.

Min-Max Normalization

$$\Rightarrow x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Original
Value (n)

Calculation

Normalized
Value

200

$$\frac{200 - 200}{800}$$

0.000

300

$$\frac{300 - 200}{800}$$

0.125

400

$$\frac{400 - 200}{800}$$

= 0.250

600

$$\frac{(600 - 200)}{800}$$

= 0.500

1000

$$\frac{1000 - 200}{800}$$

= 1.000

0, 0.125, 0.250, 0.500, 1.000

b) Z score Normalization

$$Z = \frac{x - \mu}{\sigma}$$

$$= \frac{200 + 300 + 400 + 600 + 1000}{5}$$

5

$$= \frac{2500}{5}$$

$$= 500 / \text{norm} - (x)_{\text{norm}}$$

<u>n</u>	<u>n - μ</u>	<u>$(n - \mu)^2$</u>
200	- 300	90,000
300	- 200	40,000
400	- 100	10,000
600	100	10,000
1000	500	250,000

$$\sum (n - \mu)^2 = 400\,000$$

$$\sigma^2 = \frac{400\,000}{5} = 80,000$$

$$\sigma = \sqrt{80\,000} = 282.84. //$$